Experiment ‘p_3_1_classify_int’ Results

June 3, 2026

Experiment outcome: SUCCESS
Bad responses: 0
Responses containing `assume` : 0
Responses containing `expect` : 0
Resolution attempts: 1
Hard fails (resolution): 0
Soft fails (resolution): 0
Verification attempts: 1

Problem Specification

Problem name: p_3_1_classify_int
Natural language statement: Write a method that takes an integer and returns whether it is negative, zero, or positive.
Method signature: p_3_1_classify_int (x: int) returns (s: string)

Ensures

- `x == 0 ==> s == "zero"`
- `x > 0 ==> s == "positive"`
- `x < 0 ==> s == "negative"`

GenAI interactions

Below you will find all interactions between the 'user' (program) and the 'assistant' (OpenAI).

Program → GenAI

```
You are given the following task to perform in Dafny:

Write a method that takes an integer and returns whether it is negative, zero, or
    ↪ positive.

The signature should be:

p_3_1_classify_int (x: int) returns (s: string)

The method should respect the following contract:

ensures x == 0 ==> s == "zero", ensures x > 0 ==> s == "positive", ensures x < 0 ==> s ==
    ↪  "negative"

Produce and show only the Dafny body of this method, including the curly braces that
    ↪ surround it. Do not show the signature nor contract. You must not use 'assume' nor
    ↪  'expect' anywhere in your code.
```

GenAI → Program

System fingerprint: null
ID: chatcmpl-DmdjR98ZBhumS1axev4dvclHUP05o
Model: gpt-5.5-2026-04-23
Created at: 1780484845
Finish reason: stop
Usage: 320 tokens in, and 136 tokens out

```
{
  if x == 0 {
    s := "zero";
  } else if x > 0 {
    s := "positive";
  } else {
    s := "negative";
  }
}
```

Final Program

```
method p_3_1_classify_int (x: int) returns (s: string)
       ensures x == 0 ==> s == "zero"
       ensures x > 0 ==> s == "positive"
       ensures x < 0 ==> s == "negative"
{
```

```
  if x == 0 {
    s := "zero";
  } else if x > 0 {
    s := "positive";
  } else {
    s := "negative";
  }
}
```

Total Token Usage

Input tokens: 320
Output tokens: 136
Reasoning tokens: 66
Sum of 'total tokens': 456

Experiment Timings

Overall Experiment started at 1780484844575, ended at 1780484852890, lasting 8315ms (8.32 seconds)
Iteration #1 started at 1780484844580, ended at 1780484852890, lasting 8310ms (8.31 seconds)